SAMPLE

Famous Astronomers: Ptolemy

The Geocentric Universe

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Free sample: Introduction & Chapter 1

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Introduction

Claudius Ptolemy worked in Alexandria, in Roman Egypt, in the middle of the second century, and he wrote the textbooks that the world used for the next fourteen hundred years. His great work on the heavens, which his successors eventually called the Almagest, assembled the whole astronomy of antiquity into one mathematical system: a central, motionless Earth; the sun, moon, and five planets riding their nested circles around it; and a star catalog of just over a thousand entries that fixed the sky's constellations for every later age. His book on geography gave the ancient world its coordinates and its maps; his book on astrology kept the art respectable for a millennium; and his optical work on refraction stood uncorrected until the seventeenth century.

No other single mind held the science of the West, the Islamic world, and the Renaissance under one set of instructions for so long, and the fact that the instructions were wrong in their central article, the arrangement of the heavens, only makes the story this volume has to tell the more remarkable.

The series' reader should be given the honesty this subject demands at the start: Ptolemy is a biography written in a hall of mirrors. Of the man himself almost nothing is known, not his birth, his death, his teacher, or his face; every personal fact offered in the popular accounts is either inference or invention, and one of the standard confusions, the mixing of the astronomer with the Greek kings of Egypt who shared his name, has wasted more ink than any other error in the history of science biography. This volume keeps to what the record holds: the works, their dated observations, their intellectual ancestors, and their enormous afterlife, and it says plainly, wherever the record runs out, that it has.

What the record holds is still one of the great lives in the history of science. The Almagest is the first surviving work that treats astronomy the way the modern sciences treat themselves, as a mathematical model compelled to fit measurements, tested against observed positions, and required to predict; the system it built was complicated, by the honest engineering of circles upon circles, and it predicted the planets to the accuracy the unaided eye could demand, which was the accuracy that mattered for fourteen centuries. The Geography tabulated some eight thousand places by latitude and longitude, from Britain to the Ganges, and gave the mapmaker the grammar every later map spoke.

The Tetrabiblos made astrology a subject with rules, and the Optics traced the bending of light in ways the telescope age would need. The chapters ahead treat each of these in the series' order: the life the record allows, the ideas the works contain, the tests they passed and failed, and the legacy that ran from

the Arabic astronomers through Copernicus to the death of the geocentric world.

The volume's standing irony should be named now, because it organizes everything behind it: the man who held the West's picture of the sky together longest is the man whose name is attached to the picture's fall, and the phrase Ptolemaic has come to mean both the longest-lived scientific synthesis ever made and the standard example of an elegant, useful, false theory. The reader who wants to know how science actually proceeds, how a wrong model can be built with complete honesty and immense skill, and how the truth eventually dismantles such a structure, will find no better case than the astronomer of Alexandria, and the biography that follows is the case's own story.

A word about the evidence opens the account, since the reader deserves to know what the biography rests on. The works themselves are the primary record, and their Greek and Arabic manuscript traditions, with their dates and variants, are the scholars' bedrock; the dated observations inside the works fix the career's span, as the first chapter has said; the later commentators, from the fourth century's mathematicians to the Byzantine scholars who preserved the texts, add the second layer, the tradition's own memory of its authority; and the modern scholarship, two centuries of editions and studies, stands behind every statement this volume makes about the man's era, methods, and legacy. What the sources do not hold, the volume says so: no personal letter survives, no contemporary notice, no tombstone, and the hall's mirrors are the works and their readers, which is the biography this volume honestly writes.

The city deserves a fuller paragraph than the summary gave, since Alexandria is the biography's only home. The first Ptolemy's capital, the greatest of the Hellenistic foundations, had been the ancient science's address for four centuries by the time the astronomer worked: the Library and the museum of the kings, Euclid's school, Eratosthenes' geography and geodesy, Hipparchus's generation's correspondence, and the medical and mathematical traditions that the Roman peace preserved; the city's Greek-speaking professional class, its scribes and engineers, its teachers and tax collectors, was the stratum the works' voice implies, and the works' calm reference to the courts' calendar and the provincial year tells the reader, without saying it, that the author was a working member of that class in the empire's most learned city. The reader who wants the man's milieu should

hold the picture: the museum's successors, the library's remnant, the harbor's lighthouse, and the mathematical school, in the empire's second city, under the Roman peace, in the age of the emperors who loved the learned, and the whole biography's stage is that one paragraph.

Chapter 1: The Discovery of Ptolemy

The discovery, in the series' sense, is the slow one: the recovery of the man's work by the civilizations that followed, across a thousand years and three languages, and the chapter begins with the facts of the life before turning to the strange biography of the books themselves.

The life's dates come from the work's own contents, and the scholars read them as follows. The observations the Almagest uses run from the year 127 to the year 141 of the common era, which places the active career in the reigns of the emperors Hadrian and Antoninus Pius; the geographical works and the later treatises extend the working life into the 150s and perhaps the 160s, and the traditional dating, which this volume uses with the caution the scholars use, gives the years roughly 100 to 170. The city was Alexandria, the capital of Roman Egypt, the largest Greek city of the Mediterranean after Rome itself, and the home of the great Library, whose museum and scholarly tradition, though past its golden centuries, still made the city the science's natural address; the Roman order kept Egypt's grain moving and kept the intellectual life of the Greek provinces assembled enough for a work of the Almagest's scale to be written, checked, and copied.

The name needs its paragraph, because the name is where the biography's famous confusion begins. Claudius Ptolemaeus, in the Greek of his own texts, is Klaudios Ptolemaios; the personal name Claudius has suggested to some scholars a Roman citizenship granted to a Greek family under the emperor Claudius, a generation earlier, which is plausible and unprovable; and the family name Ptolemaios was the name of the Macedonian Greek dynasty that had ruled Egypt for three centuries before the Roman conquest. He was almost certainly no kin to the kings: the name was common in Egypt precisely because the dynasty had been, and the ancient sources themselves never connect the astronomer to the royal house. The confusion is ancient, it is persistent, and it has produced a minor literature of its own; the chapters ahead return to it in the trivia, because a Famous Astronomers volume has an obligation to say plainly which Ptolemy it means.

The man's own testimony is entirely professional. The Almagest's voice is that of a working mathematician speaking to other working mathematicians: observations are cited with their dates, predecessors with their findings, models with their errors, and nowhere in the surviving work does the author step forward to say anything of his home, his family, or his life; the only self-disclosures are the practical ones, that he made certain observations himself, that others he took from the record of the earlier master Hipparchus, and that his purpose was to complete and order what the tradition had left unfinished.

Later antiquity knew little more: the commentators of the fourth and fifth centuries, writing in the same Alexandria, treat him as the authority and not as a remembered personality, and the ancient biographical tradition that preserved the anecdotes of philosophers and physicians preserved none of the astronomer.

The books' biography is fuller than the man's, and it is a story of near disappearance. The Almagest, written in Greek, copied through the later empire, passed to Byzantium as the classical tradition shrank; the geographical and minor works survived on thinner threads, and by the ninth century the Greek astronomy of the West was a set of names without texts. The rescue was the Islamic world's: the astronomers of the Abbasid Baghdad translated the great work into Arabic in the ninth century, gave it the name al-Majisti, from the Greek Megale Syntaxis, the great composition, and built a thousand years of astronomy on it, from the observatories of Baghdad and Cairo to the instrument makers of Samarkand; the Greek original itself survived in Byzantine copies, from which the modern texts descend.

The book returned to the Latin West in the twelfth century, translated from the Arabic into Latin, and the name Almagest, the Arabic article welded to the Greek title, is the scar of the journey; the West's astronomy was Ptolemaic from the moment of that translation until the death of Copernicus, and the discovery of Ptolemy by the moderns, the critical editions and the historical scholarship, is largely a tale of the last two centuries, which is why the biography of the man is still being written, and still being freed from the errors of the earlier books.

One further discovery belongs in the chapter, the series' usual closing item: the identity of the works. The Almagest stands first. The Geography, eight books of coordinates and mapmaking method, stands second, and its survival, through a single damaged manuscript tradition, is itself a romance. The Tetrabiblos, four books on astrology, is the third great pillar; the Optics, the Planetary Hypotheses, the Handy Tables, the Analemma, the Harmonics, and a handful of minor pieces complete the list, and the modern scholarship that authenticated, dated, and ordered this library was the work of generations of philologists, which means the scientific discovery of Ptolemy, as the

chapters ahead keep showing, was itself a science, conducted with the same patience as the astronomy it recovered.

A note on the words the chapters use, offered once and then trusted. The deferent is the main circle, the epicycle the rider circle, the eccentric the offset of the main circle's center from the Earth, and the equant the point from which the riding is uniform; the anomaly is a planet's misbehavior, the retrograde

loop or the speed's variation, from the Greek word for irregularity; and the save the appearances of the older histories is the ancient task's motto, the command to reproduce the sky's show, whatever the machinery's cost in elegance. The reader who keeps those six terms holds the whole system's vocabulary, and the chapters ahead use no mathematics beyond them.

One more idea belongs in the set before the chapter closes, the system's economy, since the reader should see how much the machine accomplished with how little. The whole planetary apparatus, for all its epicycles and equants, uses only the one curve and the one speed, the circle and the uniform motion, repeated in combination; the parameters number in the few dozens, the radii and rates of the wheels, against the thousands of observations the system reproduces; and the economy's lesson, which the historians of science teach from the case, is the standing one about models, that a wrong mechanism can still be an efficient summary of the data, and that the summary's economy is a real scientific virtue even in a false theory. The idea's modern descendant is the observer's standing caution, that a model's fit is not the truth's proof, and the Almagest is the caution's founding case, taught in the modern classrooms from the original text.

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